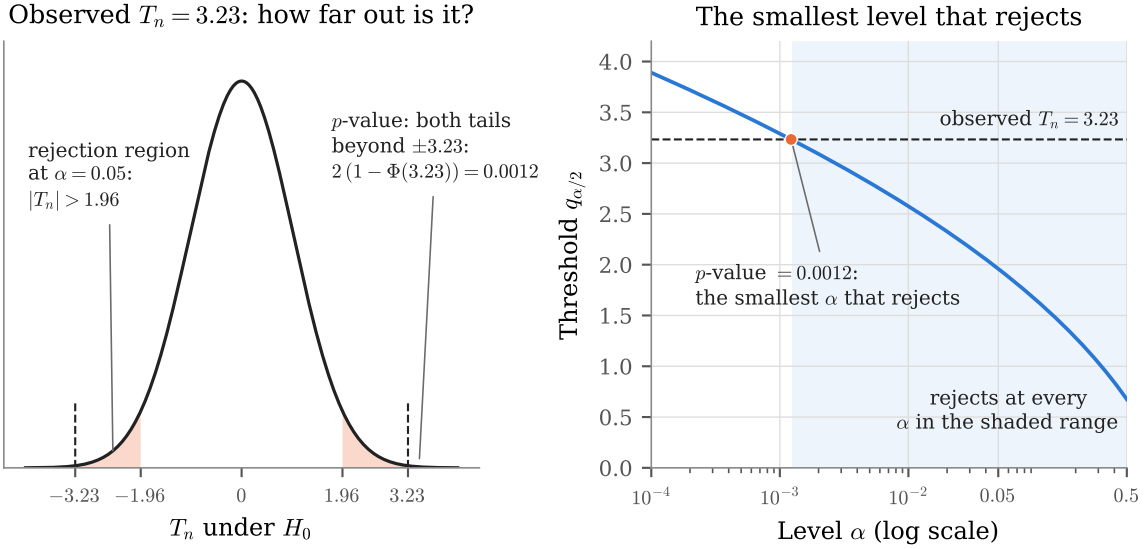
Observed $T_n = 3.23$: how far out is it?
rejection region
at $\alpha = 0.05$:
$|T_n| > 1.96$
p-value: both tails
beyond ± 3.23:
$2\,(1 - \Phi(3.23)) = 0.0012$
−3.23
−1.96
0
1.96
3.23
T_n under H_0
The smallest level that rejects
Threshold $q_{\alpha/2}$
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
observed $T_n = 3.23$
p-value $= 0.0012$:
the smallest α that rejects
rejects at every
α in the shaded range
10^{-4}
10^{-3}
10^{-2}
0.05
0.5
Level α (log scale)